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AND PHYSICS

PROJECT PLAN

**A molecular dynamics investigation of Styrene
maleic acid (SMA) polymers and how it
interacts with Lipid bilayer under different
biological environments**



3 October 2018



Scientific Report in the 3rd Year Module MTH3005M
Mathematics Project





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chapter and section should be numbered.

Introduction

Styrene maleic acid (SMA) is a transmembrane protein. These types of proteins are important to help understand the functionality of cell membranes, which can be applied to future applications in nanotechnology/nanomedicine. However, before being able to apply it anywhere there are several challenges in working with transmembrane proteins such as the insolubility in water. In recent studies it has shown that SMA polymers can play the role of traditional detergents to isolate proteins directly from cell membranes forming SMA lipid particles which retain the lipid environment of membrane proteins.

→ References are missing.

SMALP consists of a polymer forming a ring around a patch of membrane protein with surrounding phospholipids, therefore creating a disc-like structure. The protein is encapsulated from the cell membrane. This retains the lipid environment and therefore its structure won't be destroyed while using detergents. The problem is that the exact formation route and structure of the polymer surrounding the disc is unknown.

The objective of this project is to do experiments and molecular dynamic simulations to characterise the SMA, and investigate using molecular dynamics how the polymer interacts in different biological environments and with the lipids.

very good : Introduction of the system with the challenges.

Motivations

↳ you could here combined this section with the previous one.

Physics plays a huge part of medical science and being able to embark with extremely new research is always a pleasure. With AS/A2 knowledge of biology and chemistry undergoing this project will combine all of my prior knowledge.

Previous work that has been done in the previous studies.

This project overlaps a couple of previous works that I have done before in the past. In AS biology I have studied transmembrane proteins, lipids and about the cell membrane. In A2 chemistry I did polymers and molecular structure of amino acids/proteins. With just these two it greatly helps me understand structures of cells and proteins on a molecular level, which can also be applied to understanding new structures.

Within condensed matter physics which I did in the second year, there was a topic called 'Amphiphiles'. From that topic I calculated the shape, density & size etc... which will aid me in doing the same to SMA polymer, and how its assembly can be optimised.

It is good to add how it is related to your previous studies.



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Literature Survey

Author	Year	System
Stephen C. L. Hall, Cecilia Tognoloni, Gareth J. Price, Bert Klumperman, Karen J. Edler, Tim R. Daffern, and Thomas Arnold	2018	SCMA at pH 8, SMA synthesized in one step by mixing initial reagents of styrene and maleic anhydride in a 2.34:1 molar ratio.
Jana Broecker	2017	Used SMA copolymers
Rodrigo Cuevas Arenas, Johannes Klingler, Carolyn Vargas and Sandro Keller	2016	SMA copolymers directly solubilise phospholipids and membrane proteins from artificial and natural bilayers to form SMALP.

It is not clear how some references are related to your project/report.

Literature Review

List of references

- [1] S. C. L. Hall, C. Tognoloni, G. J. Price, B. Klumperman, K. J. Edler, T. R. Daffern, and T. Arnold, "Influence of Poly(styrene-co-maleic acid) Copolymer Structure on the Properties and Self-Assembly of SMALP Nanodiscs," *Biomacromolecules*, vol.19, pp. 761-772, 2018.
- [2] J. Broecker, "Crystallogenesis of Membrane Proteins Mediated by Polymer-Bounded Lipid Nanodiscs," *ResearchGate*, vol. 25, pp. 1-9, Feb. 2017. [Online] Available: <http://dx.doi.org/10.1016/j.str.2016.12.004>
- [3] R. C. Arenas, J. Klingler, C. Vargas and S. Keller, "Influence of lipid bilayer properties on nanodisc formation mediated by styrene/maleic acid copolymers," *Royal Society of Chemistry*, vol. 8, pp. 1-11, June. 2016. [Online] Available: www.rsc.org/nanoscale ✓
- [4] Y. Fukuda, K. Miyamae and Y. Sasanuma, "Computational design of polymers: poly(ester amide) and polyurethane," *Royal Society of Chemistry*, vol. 7, pp. 1-12, Jul. 2017. ✓



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Equipment & Facilities

Good.

- Computer (able to run simulations) ✓
- Only running cost is electricity of the room and the computer ✓

Action Plan

Task Name	Duration	Start	Finish	Predecessors	A	Oct					Nov					Dec 2	De
						3	Sep 30	Oct 7	Oct 14	Oct 21	Oct 28	Nov 4	Nov 11	Nov 18	Nov 25		
Research Plan	11d	09/30/18	10/12/18														
Logbook 1	10d	10/12/18	10/25/18														
Logbook 2	22d	10/25/18	11/23/18														
Progress report	1d	12/06/18	12/06/18														
Logbook 3		11/23/18	TBC														
Scientific Report		TBC	TBC														
Viva		TBC	TBC														

*The tasks consist in what you will do
learning a new software for example.*

Meeting times

Dates	Time (hours)
15/10/18	1100
22/10/18	1100
29/10/18	1100
5/11/18	1100 ✓
12/11/18	1100
19/11/18	1100
26/11/18	1100

Captions are missing.

